

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Ritam Santra
Roll Number	MDE2026009@iiita.ac.in
Programme / Department	Information and Technology
Topic ID	19
Full Assigned Topic	An Automated Pipeline for Flagging Outdated Citations in AI-Recommended Literature
AI Tool Used	Gemini, chatgpt, claude
Date	21/08/26

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☒ Almost daily

A2. Have you previously asked an AI system to generate references?

☒ Frequently ☐ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☒ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Gemini, Chatgpt
Model name	Gemini Pro
Model/version shown	1.5 Pro
Access tier	☐ Free ☒ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☒ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☒ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☒ Yes ☐ No
Date/time generated	

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title:

Measure	Value
Approximate pages	8
Approximate word count	3450
Number of sections	10
Total number of references	10
Approximate in-text citation occurrences	28
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	1
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	20
URL only	20
No persistent identifier supplied	0
TOTAL	41

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	10
R5	18
R6	26
R7	34
R8	39
R9	40
R10	41

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure

4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Valid survey paper title on citation recommendation with official DNB link.	5	Yes
R2	Matches R1 title, hosted on ResearchGate.	5	Yes
R3	Real paper title on BERT embeddings with specific arXiv ID	5	Yes
R4	Highly specific technical title on claim networks with valid arXiv ID	5	Yes
R5	ACL Anthology URL (2023.emnlp-main.640.pdf) matching standard venue naming	5	Yes
R6	Truncated arXiv title on SemanticCite system with valid arXiv ID	4	Yes
R7	Complete Emerald publisher DOI for AI retraction study.	5	Yes
R8	Specialized IJERT multi-agent literature research paper.	5	Yes
R9	Detailed arXiv preprint on patent-paper knowledge flow (2512.24259v1).	5	Yes
R10	Standard arXiv paper on reviewer assignment benchmark (2303.16750v2).	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
Verified 10/10 papers exist on official host platforms (arXiv, Emerald Insight, ACL Anthology, IJERT, DNB).
- Is the title correct?
8/10 titles matched perfectly; 2/10 exhibited AI web-scraping truncations or header formatting artifacts (R5, R6).
- Are the authors correct?
8/10 matched completely; 2/10 omitted author metadata entirely in the raw output (R1, R2).
- Is the year correct?
10/10 matched official publication or preprint submission dates.
- Is the journal/conference/source correct?
10/10 accurately pointed to the correct conference, journal, or preprint repository.

- Are volume, issue and pages correct, if supplied?
Verified against official publisher records where supplied (e.g., Emerald Insight for R7).
- Does the DOI/arXiv/PMID belong to this publication?

10/10 DOIs, arXiv IDs, and target URLs belonged exclusively to the matched papers.

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID Direct resolution via doi.org (R7) and arxiv.org/abs/ (R3, R4, R6, R9, R10).
- ☐ Publisher or official repository ACL Anthology (aclanthology.org) for R5 and IJERT repository for R8.
- ☐ Google Scholar Crossref metadata index for DOI verification; Deutsche Nationalbibliothek (DNB) for R1.
- ☐ Semantic Scholar / OpenAlex Google Scholar exact-title and author-keyword searches to locate primary landing pages and verify missing author credits for R1 and R2.
- ☐ Crossref Queried Crossref's official DOI registry for R7 ([10.1108/GKMC-09-2024-0582](https://doi.org/10.1108/GKMC-09-2024-0582)) to verify the publisher record, author list, and journal details in *Global Knowledge, Memory and Communication*.
- ☐ PubMed or another discipline-specific database Used arXiv, ACL Anthology, and IJERT repositories as primary CS/AI discipline databases to verify preprints and proceedings for R3–R6 and R8–R10. (No PubMed entries applied to this dataset).
- ☐ Exact-title search Executed verbatim title searches across Google Scholar and Semantic Scholar for R1, R2, R5, R6, and R8 to locate landing pages and identify title truncations or web-scraping artifacts.
- ☐ Author + title-keyword search Performed targeted keyword queries for R1 and R2 to retrieve the missing primary authors (Färber & Jatowt, 2020) omitted by the AI's raw output.

Not acceptable as sole evidence: "I asked another AI and it said the paper exists." AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	N	Y	Y	Y	Y
R6	Y	N	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	Google Scholar / DNB	URL target	Paper exists (Färber & Jatowt, 2020); omitted author names in AI output.
R2	ResearchGate	URL target	Duplicate ResearchGate entry for Färber & Jatowt (2020); lacks standard author metadata.
R3	arXiv Repository	arXiv:2405.04136v1o	Verified exact title and authors (Wolff et al., 2024).
R4	arXiv Repository	arXiv:2605.30966	Verified exact title and authors (Jia et al., 2026).
R5	ACL Anthology	2023.emnlp-main.640	Paper exists (Rohatgi et al., 2023); AI truncated title ("Computational" vs "Linguistics").
R6	arXiv Repository	arXiv:2511.16198v1	SemanticCite paper exists (Sebastian Haan, 2025); formatting artifacts present in title.
R7	Emerald Insight / DOI	10.1108/GKMC-09-2024-0582	Exact match for published journal article in GKMC journal.
R8	IJERT Repository	URL target	Verified publication on autonomous multi-agent literature research.
R9	arXiv Repository	arXiv:2512.24259v1	Verified exact title and authors (Rose et al., 2025).
R10	arXiv Repository	arXiv:2303.16750v2arXiv Repository	Verified standard benchmark paper for reviewer assignment problem.

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	/B	Publication exists, but AI omitted author names from bibliography record.
R2	A	Paper exists on ResearchGate, but lacks standard author metadata.
R3	A	Publication exists and all bibliographic metadata and persistent identifier match.
R4	A	Publication exists and all bibliographic metadata match official arXiv record.
R5	A	Article exists, but AI truncated the title string in output bibliography.

R6	A	Article exists, but title contains leading formatting artifacts and truncated text.
R7	A	Fully verified against official journal DOI and publisher database.
R8	A	Fully verified against arXiv repository with exact title and author match.
R9	A	Fully verified against arXiv repository with exact title and author match.
R10	D	

Verified standard benchmark paper for reviewer assignment problem.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 6 B = 4 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(\underline{6}) + 3(\underline{4}) + 1(\underline{0}) + 0(\underline{0}) + 1(\underline{0}) = \underline{36}$ points

Authenticity Score = $\underline{36} / (4 \times \underline{10}) \times 100 = \underline{90.0} / 100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = $4(6)+3(2)+1(1)=31$; maximum = 40; Authenticity Score = $31/40 \times 100 = 77.5$.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 90 Total predictions = 100 Prediction Accuracy = 90 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	41
References deeply audited	10
A: Verified	6
B: Wrong metadata	4
C: Frankenstein	
D: Fabricated	
E: Identifier mismatch	
Authenticity Score	/100
Prediction Accuracy	%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

.....Yes, in this case, enabling active web search and retrieval features allowed the model to retrieve genuine publications. However, visual formatting can be deceptive: references that appeared incomplete or clipped still pointed to authentic preprints and published papers.....

.....

2. Did all genuine references actually support the claims for which they were cited?

.....Yes. When checked against the in-text citation contexts, the cited works directly supported the specific technical claims made in the manuscript (e.g., SemanticCite for full-text NLI verification, SPECTER2 for dense document embeddings, and OpenAlex for retraction monitoring).....

.....

3. What was the most serious citation-integrity problem you observed?

.....Metadata truncation and web-scraping artifacts. Instead of producing standard APA/IEEE bibliographic citations with complete author lists, the model pasted scraped web page titles (e.g., missing authors, truncated titles ending in ..., or raw URLs), requiring manual cleanup before academic publication.....

.....

4. What is the single most important lesson you learned from this lab?

.....Web-augmented AI models can ground content in real scientific papers, but they frequently output malformed metadata. Researchers must verify both paper existence and the completeness of the bibliographic record.....

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SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_MDE2026009_RITAM_Santra

Example: Lab1_MDE2026009_Ritam_Santra

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name:ritam santra.....

Roll Number:MDE2026009.....

Signature:Ritam Santra.....

Date:21/08/2026.....

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
Verification -> Quantitative Scores -> Submission